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Supplemental Instructions for Focus Adjustment of the Acton Focusing Mirror

Due to slight variations in the distance from the mounting flange of the Detector to its focal plane, it may be necessary to adjust the focusing mirror in the Acton 300i to obtain optimum resolution.

The following procedure is recommended for optimizing focus (if required):

1. Locate the focal Plane Adjustment Micrometer (shipped separately) and the Mirror Mounting Spindle, as shown in figure 1.
2. Assure the Micrometer Spindle is not protruding beyond the end of the Mirror Mount and attached the Mounting Block to the Mirror Mounting Spindle using the #8-32 Mounting Screw supplied.
3. Rotate the Micrometer Thimble gently until the Micrometer Spindle just contacts the Mirror Mount. Do not apply excess force. Note and record the micrometer setting (factory setting).
4. It is recommended that focus adjustments be made in 0.003 inch increments, equivalent to 3 divisions on the Micrometer. The total adjustment required should not exceed 0.080 inches.
5. To make a focus adjustment, remove the Acton 300i cover and loosen the two (2) Set Screws securing the Mirror Mounting Spindle (approximately 1/8 turn). Gently push on the end of the Mounting block while turning the Micrometer Thimble clockwise 0.003 inches, equivalent to 3 divisions on the Micrometer. Assure the Micrometer Spindle contacts the Mirror Mount and re-tighten both Set Screws. To assure repeatable results, gently rotate the Mirror Mounting Spindle slightly clockwise and counterclockwise while tightening the Set Screws. This assures that the Set Screws will align correctly with the machined flat surface of the Mirror Mounting Spindle.
6. Replace the Acton 300i cover and check the resolution. If the instrument is still out of focus, repeat the procedure and adjust focus 0.003 inches, equivalent to 3 divisions on the Micrometer, in the same direction and recheck resolution. If resolution improves, continue moving in the same

direction until optimum resolution is achieved. If resolution degrades, adjust for focus in the opposite direction until optimum resolution is achieved.

NOTE: Total movement should not exceed the original setting (as recorded in Step 3) by more than +/- 0.080 inches.

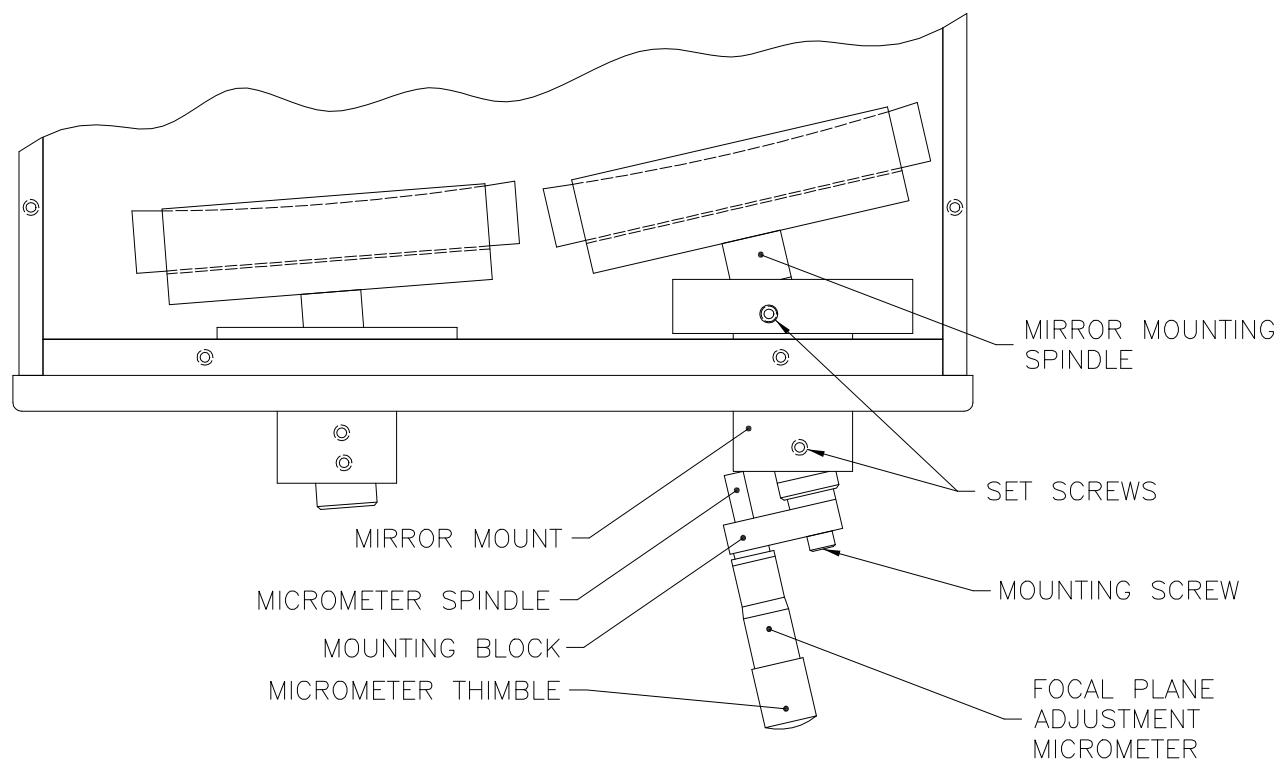


Figure 1